

Air Quality Index (AQI) Prediction & Analysis — India

Project Overview

Air pollution is one of the most critical environmental challenges affecting public health and quality of life. This project focuses on analyzing historical air quality data from major Indian cities and developing an intelligent system to understand pollution patterns and forecast future Air Quality Index (AQI) levels.

The project combines data analysis, feature engineering, machine learning, and interactive visualization to provide meaningful insights into air pollution trends across India.

Objectives

- Analyze historical air quality data from Indian cities.
- Identify key pollutants affecting AQI.
- Study seasonal and city-wise pollution patterns.
- Discover long-term AQI trends.
- Predict future AQI values using machine learning.
- Build an interactive dashboard for visualization and prediction.

Dataset Information

Source

Central Pollution Control Board (CPCB) Air Quality Dataset

Coverage

- Multiple Indian cities
- Daily air quality measurements
- Several years of historical records

Key Attributes

Location Information

- City
- Monitoring Station
- Date

Pollutant Measurements

- PM2.5
- PM10
- NO₂
- SO₂
- CO
- O₃
- Benzene
- Other air quality indicators

Target Variable

- Air Quality Index (AQI)

Exploratory Data Analysis (EDA)

The analysis phase focuses on understanding the dataset and extracting meaningful information.

Performed Analysis

- Dataset structure inspection
- Missing value analysis
- AQI distribution analysis
- City-wise AQI comparison
- Pollutant distribution analysis
- Correlation analysis
- Monthly AQI trends
- Seasonal AQI patterns
- Yearly AQI trends
- Pollutant impact assessment

Key Visualizations

- Histograms
- Boxplots
- Line Charts

- Heatmaps
 - Correlation Matrices
 - Trend Analysis Graphs
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Data Preprocessing

The dataset was cleaned and transformed to improve model performance.

Preprocessing Tasks

- Missing value treatment
- Date formatting
- Feature extraction
- Outlier inspection
- Dataset cleaning
- Feature generation

Engineered Features

Time-Based Features

- Month
- Day of Week
- Quarter
- Season

Historical AQI Features

- Previous Day AQI
- 3-Day AQI Lag
- 7-Day AQI Lag

Rolling Statistics

- 3-Day Moving Average
- 7-Day Moving Average
- 30-Day Moving Average

These features help capture temporal pollution patterns and improve forecasting accuracy.

Machine Learning Approach

Multiple machine learning models were evaluated for AQI prediction.

Models Evaluated

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

Evaluation Metrics

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R² Score

The best-performing model was selected for deployment in the dashboard.

Key Insights

1. Strong Seasonal Impact

AQI levels increase significantly during winter months due to:

- Low wind speed
 - Temperature inversion
 - Pollutant accumulation
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2. PM2.5 is the Most Influential Pollutant

Analysis indicates that PM2.5 has a strong relationship with AQI levels and is one of the primary contributors to poor air quality.

3. Major Metropolitan Cities Show Higher AQI

Large urban centers consistently experience higher AQI values due to:

- Dense traffic
- Industrial activities
- Population concentration

4. AQI Exhibits Temporal Patterns

Historical AQI values play a significant role in forecasting future AQI, indicating strong time-series behavior.

5. Air Quality Varies Across Regions

Different cities exhibit distinct pollution characteristics, emphasizing the need for localized pollution control strategies.

Interactive Dashboard Features

The project includes a Streamlit-based dashboard with:

Dashboard Components

- City Selection
- AQI Trend Analysis
- Monthly Pollution Analysis
- Pollutant Breakdown
- City Comparison
- AQI Category Classification
- Future AQI Prediction

AQI Categories

AQI Range	Category
0 – 50	Good
51 – 100	Satisfactory
101 – 200	Moderate
201 – 300	Poor
301 – 400	Very Poor
401+	Severe

Future Enhancements

- Real-time AQI monitoring
 - Weather data integration
 - Deep Learning models (LSTM/GRU)
 - Forecasting for multiple future days
 - Cloud deployment
 - Automated data updates
 - Explainable AI integration
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Technologies Used

Data Analysis

- Python
- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn
- Plotly

Machine Learning

- Scikit-Learn
- XGBoost

Deployment

- Streamlit
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Author

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